



MOTIVATION



Humans often reveal their next move through implicit social cues.

- Body posture, orientation, and gait provide rich cues about where and how a person is likely to move, which are critical for socially compliant navigation.
- But existing methods often explicitly predict these cues before planning.
- Can a robot navigate socially **without explicitly predicting social cues**?

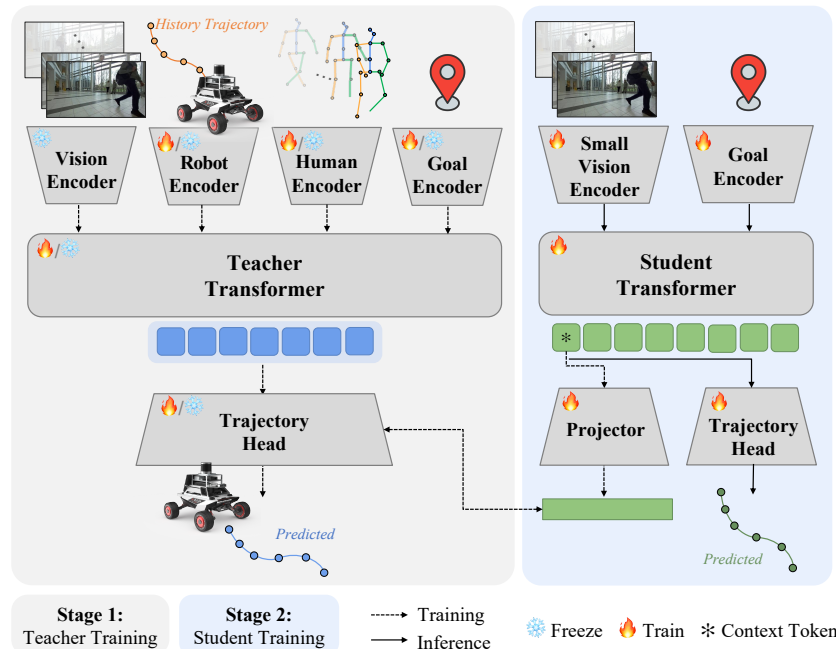
KEY IDEA

*Learn **explicit** cues. Navigate with **implicit** awareness.*

- HumAIN uses privileged skeletal cues during training to learn human-aware representations, then distills this knowledge into a visual navigation policy.

TWO-STAGE TEACHER-STUDENT FRAMEWORK

No skeleton. No explicit human-state estimation at deployment.



Stage 1 • Social-Cue-Aware Teacher

- Learns trajectory prediction from vision, robot history, skeletal keypoints, and goal.
- Privileged skeletal cues provide explicit knowledge of human posture and motion.

Stage 2 • Implicit Student

- Observes only vision and goal while learning from the frozen teacher.
- Distills the teacher's human-aware latent representation through feature alignment.
- Predicts socially compliant trajectories without explicit human-state estimation at deployment.

Together, HumAIN distills explicit social knowledge into implicit visual representations for socially compliant navigation.

RESULTS



- **Improved Social Compliance:** HumAIN generates more socially compliant trajectories across diverse indoor and outdoor human-centered environments.
- **Better Human Awareness:** Predicted trajectories adapt to subtle human behaviors, including body posture, orientation, and gait.
- **Benefits of Privileged Social Cues:** Ablations show that incorporating skeletal keypoints during teacher training improves socially aware trajectory prediction.
- **Effective Knowledge Distillation:** Distilling the teacher's human-aware representation enables the student to retain social awareness using only visual and goal inputs at deployment.